

REVIEW

A Review on Multi-Types Analysis for Graph Neural Networks-Based Recommendation Systems

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Abstract

Recommender systems (RS) aim to suggest relevant products such as items, films, and music based on user preferences, and have long been a focus in both industry and academia. While traditional RS approaches, including collaborative filtering (CF) and content-based methods, have achieved considerable success, they often struggle with challenges such as data sparsity and limited modeling of complex relations. Recently, graph neural networks (GNNs) have emerged as powerful tools for RS by capturing rich structural and semantic dependencies in user-item interactions. This paper presents a comprehensive survey of GNN-based RS with a particular focus on multi-type scenarios, where multiple behaviors, data modalities, or graph relations coexist. We first introduce a clear taxonomy that distinguishes multi-types in RS and classify existing GNN-based models according to graph construction strategies, embedding learning mechanisms, and application contexts. We then analyze the motivations for adopting GNNs in multi-type RS and review representative methods, highlighting how they address challenges in framework design, embedding layer construction, and computational efficiency. Finally, we discuss open issues such as scalability, interpretability, and dynamic graph modeling, and outline potential research directions to guide future advancements in this field.

Keywords Multi-types · Recommendation systems · Graph neural networks · Collaborative filtering · Future directions

1 Introduction

Recommender systems (RS) are widely regarded as an advanced network technology that leverages data mining, natural language processing, and domain knowledge to identify and recommend products or information that best match user preferences. By analyzing factors such as historical user behavior and social relationships [1], RS provide personalized and convenient services for consumers while generating substantial benefits for enterprises [2]. A widely recognized assertion in this field is that information overload for users remains a critical challenge. In other words, RS aim to address the problem of matching users with relevant information entities by establishing effective connections between users and the content of interest, thereby fulfilling user needs [3]. As a result, many companies have prioritized the development and deployment of RS applications. For example, a large proportion



of purchased items on Amazon and other e-commerce platforms are the result of recommendations generated by RS [3, 4].

In recent years, deep learning has attracted widespread attention [5–7], in particular, graph neural networks (GNNs) have had a significant impact on various domains [8–10]. By revolutionizing artificial intelligence, GNNs have created new opportunities that can also benefit research on recommender systems (RS). Specifically, GNNs offer effective solutions to long-standing challenges in traditional RS, such as the cold-start problem and data sparsity, thereby fostering the development of more advanced methods. On the one hand, GNNs can automatically learn representative features from multi-source heterogeneous data. By mapping different data types into the same latent space, they enable the generation of unified representations. On the other hand, GNNs can model the large-scale, complex relationships between users and items by learning deep, nonlinear network structures [11]. This capability allows GNNs to capture essential characteristics from datasets and derive deep feature representations of users and items [12]. Owing to these advantages, GNN-based RS outperform traditional recommendation algorithms in mitigating data sparsity and cold-start issues, making them a research hotspot in both academia and industry. In 2016, ACM RecSys hosted the first symposium on GNN-based RS (DLRS'16), which identified GNNs as a promising research direction for RS; DLRS'17 was subsequently held in Como, Italy. Moreover, leading conferences in data mining and machine learning (e.g., SIGKDD, NeurIPS, SIGIR, WWW, and AAAI) have reported a steady increase in the number of publications on GNN-based RS in recent years. Many universities and research institutions have also conducted extensive studies in this area [13].

With the rapid development of information technology, various aspects of recommender systems (RS) have been profoundly enhanced, particularly in terms of recommendation performance. Building on these advancements, GNNs have been applied in diverse technological contexts, such as Graph Spatial–Temporal Networks (GSTN) [14], Graph Autoencoders (GAE) [15], Graph Attention Networks (GAT) [16], Graph Sample and Aggregate (GraphSAGE) [17], and Graph Convolutional Networks (GCN) [18]. RS implemented under these different frameworks pursue distinct application objectives in their respective domains and have achieved notable improvements in recommendation performance. However, GNN-based RS employing different methodologies often demonstrate substantial variations in recommendation performance, accuracy, and diversity across application scenarios. Compared with single-behavior data, multi-behavior information can offer richer knowledge to RS, enabling the capture of more comprehensive semantic patterns. Consequently, multi-behavior modeling has attracted growing attention from researchers. For example, [19] developed a model named MBGCN that integrates multi-behavior data into a unified graph structure, while [20] proposed a multi-behavior recommendation framework based on a Graph Meta Network, incorporating a meta-learning paradigm into multi-behavior pattern modeling. Beyond traditional multi-behavior modeling, hybrid multi-behavior GNN-based RS have also emerged. For instance, [21, 22] employed a GNN model for multi-behavior session-based recommendation. The applications and details of each scenario are discussed in Sect. 4.

GNNs have become a cornerstone in the development of advanced recommender systems (RS) due to their strong capability to model high-order collaborative signals and complex user-item relationships. As the field continues to evolve rapidly, several surveys have been published to provide comprehensive overviews of GNN applications in recommendation. For example, [2, 23–25] provide a broad survey of GNN-based RS that focuses on generic architectural patterns, learning mechanisms, and application scenarios, and organize models mainly along dimensions such as recommendation stage, scenario, and objective. These works offer foundational insights into the general design space of GNN-based RS but pay limited attention to the diversity of data types and interaction patterns. More specialized surveys have subsequently emerged. For instance, contrastive learning in GNN-based recommendation [26–28] has been reviewed with a focus on training paradigms and self-supervised objectives, while heterogeneous graph-based recommendation models [29–31] have been surveyed from the perspective of multiple entity types, yet with little discussion of diverse interaction types and multi-view fusion. Research on multi-behavior recommendation (e.g., Kim et al. and related studies) has systematically analyzed behavioral heterogeneity, but it typically concentrates on multi-behavior modeling itself and provides only limited integration with GNN architectures. Other surveys independently investigate sequential GNN models [32–34], multi-modal

GNN fusion [35–37], or graph contrastive learning-based RS (GCL-RS) [38–40], each addressing one aspect of type variation in isolation. An overview of more recent literature on GNN-based RS is provided in Table 1.

However, to date, none of the existing surveys have provided a unified, type-oriented perspective that systematically categorizes GNN-based recommendation algorithms across multiple dimensions of heterogeneity [41]. In this survey, the term “type” refers to heterogeneous categories such as multi-behavior, multi-modal, and multi-relational graph structures (Fig. 1). Concretely, we focus on three tightly related dimensions: (i) interaction behaviors (e.g., clicks, favorites, purchases), (ii) data modalities (e.g., textual, visual, or spatio-temporal signals), and (iii) graph relation structures and views (e.g., user-user, user-item, item-item, item-entity, social and knowledge graphs [42], as well as sequential or spatial-temporal views).

The absence of a multi-type analysis along these dimensions limits our understanding of how different GNN variants are designed to address complex, real-world data structures in RS tasks [43]. To bridge this gap, this survey introduces a multi-type taxonomy and analytical framework that synthesize existing research through the lens of multi-type modeling. First, we formalize the notion of multi-types in RS and construct a taxonomy that jointly considers behavioral, modal, and relational types. Second, we map representative RS scenarios (e.g., multi-behavior, social, sequential, knowledge-aware, and multi-modal recommendation) onto this taxonomy, thereby clarifying which types each scenario naturally involves. Third, we analyze how different families of GNN architectures such as GCN, GraphSAGE, GAT, GAE, and GSTN are instantiated under various multi-type settings, with a focus on how they construct graphs, propagate messages, fuse heterogeneous signals, and alleviate sparsity or imbalance across types. Finally, based on this framework, we summarize cross-type challenges and outline future research directions for unified learning across diverse data forms.

In summary, the main contributions of this study are as follows:

- **Unified multi-type taxonomy for GNN-based RS.** This study proposes, to the best of our knowledge, the first survey that centers on a unified multi-type perspective for GNN-based recommender systems. We explicitly

Table 1 References on GNN-based review of RS

Literature	Description	Contributions
Wu et al. [24]	It gave an overview of how GNN are used in RS	Survey of GNN-based recommender systems; categorization, technical features, performance comparison, and future directions
Gao et al. [29]	Highlights core methods, challenges, and roadmap	Survey of GNN-based RS; categorization, key challenges, and research roadmap
Kim et al. [4]	It presented a survey on multi-behavior RS	Classifies user behaviors, reviews fusion/transfer/disentanglement techniques, and discusses evaluation and challenges
Sharma et al. [32]	Surveys GNN-based social recommendation systems	Categorizes GNN-based social recommendation, analyzes relation impact, and outlines challenges
Anand et al. [2]	The approach presented a survey on recommender systems using GNN	Reviews GNN architectures for RS, classifies models by graph construction and learning strategy, and summarizes datasets, metrics, and open issues
Li et al. [201]	A survey on GNN-based recommendation in social networks	Reviews GNN models leveraging social ties, analyzing graph construction, message passing, and challenges like sparsity and dynamic influence
Wang et al. [164]	A review of graph learning-based RS	Provides a unified view of graph learning for RS, classifies models by objectives and graph types, and outlines challenges and future directions
Amara et al. [15]	It presented a review of graph learning approaches in RS	Contributions: (1) categorize graph learning models by task, (2) summarize key techniques, and (3) discuss open issues and trends
Dang Yue, [96]	They proposed a review of RS based on graph embedding techniques	Summarized graph embedding methods for RS, classified by strategies and graph types, and discussed pros, cons, and future directions
Duricic et al. [202]	The paper presented a review on beyond-accuracy aspects in GNN-based RS	Focused on diversity, serendipity, and fairness in GNN-based RS, reviewed related models/metrics, and discussed challenges beyond accuracy
Gao et al. [203]	It presented a survey on deep learning methods over knowledge graphs for RS	Reviewed deep learning approaches integrating knowledge graphs into RS, classified by construction and embedding strategies, and discussed scalability and integration challenges
Zhou et al. [204]	They presented a comprehensive survey of deep learning-based RS	Categorized deep learning models for RS, analyzed key architectures (CNNs, RNNs, GNNs), and summarized challenges and future trends

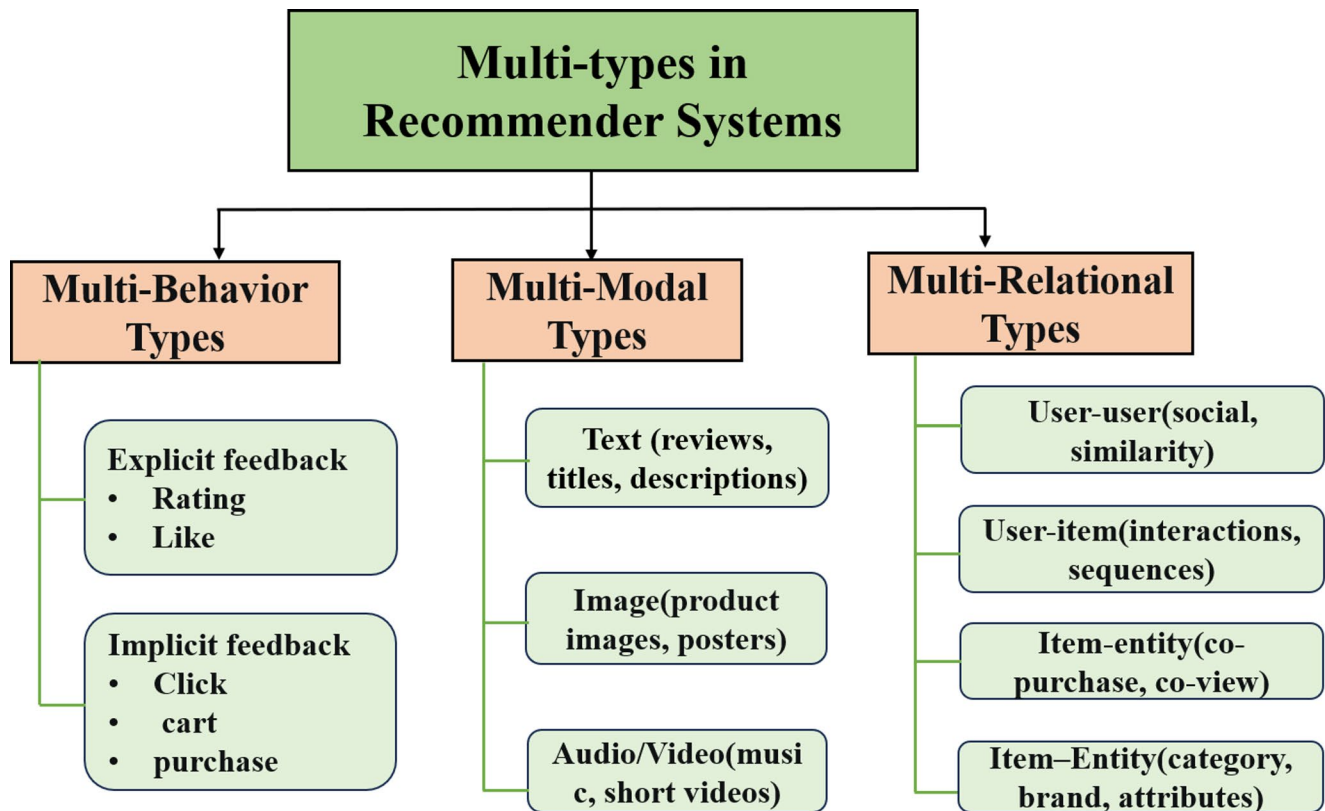


Fig. 1 A taxonomy of multi-types in RS, covering behavioral types, data modalities, and graph relation structures

define “type” in terms of multi-behavior, multi-modal, and multi-relational graph structures, and organize existing work according to this taxonomy rather than by model architecture alone.

- **Type-oriented analytical framework across RS scenarios and GNN architectures.** Building on the description of both traditional and GNN-based RS, we develop an analytical framework that connects common RS scenarios with the proposed multi-type taxonomy and different GNN architectures. This framework enables a systematic comparison of how various GNN-based models construct and exploit heterogeneous behaviors, modalities, and graph relations.
- **Cross-type synthesis of challenges and future directions.** We analyze recent multi-type RS research published in leading conferences and journals, classify the literature according to the multi-type challenges it addresses, and provide corresponding commentary. Based on this synthesis, we identify open problems specific to multi-type GNN-based RS such as cross-type sparsity, imbalance, heterophily, and multi-view fusion, and outline promising future research directions.

The remainder of this paper is organized as follows. Section 2 establishes the preliminaries and foundations, explicitly defining the multi-type taxonomy and reviewing traditional recommendation paradigms to highlight their limitations in heterogeneous scenarios. Section 3 constitutes the core analysis, systematically reviewing representative GNN architectures, including GCN, GraphSAGE, GAT, GAE, and GSTN, through the lens of multi-behavior, multi-modal, and multi-relational modeling. Section 4 discusses open problems and outlines future research perspectives, and Sect. 5 concludes the paper.

2 Preliminaries and Foundations

This section formalizes the concept of multi-types, outlines the basic recommender system paradigms, and analyzes why traditional approaches fall short in handling heterogeneous data, motivating the need for GNN-based solutions.

2.1 Definition of Multi-types in Recommender Systems

In the context of RS, the term multi-types refers to distinct forms of heterogeneity [44, 45]. To provide a unified analytical lens, we categorize multi-types into three dimensions (as illustrated in Fig. 1):

- i. Multi-behavior types: different forms of user-item interactions (e.g., click, cart, purchase) that imply varying levels of user intent.
- ii. Multi-modal types: heterogeneous content modalities associated with users or items, such as textual descriptions, visual images, or audio signals.
- iii. Multi-relational types: diverse topological structures in graphs, including user-user social links [46], item-entity knowledge associations, and spatio-temporal dependencies.

Throughout this survey, we employ this taxonomy to analyze how GNN architectures address the specific challenges posed by these heterogeneity types.

2.2 Traditional Recommender Systems

Recommender systems generally filter information to predict user preferences. While effective in early stages, traditional paradigms face intrinsic limitations when applied to the complex multi-type scenarios defined above [47–51].

2.2.1 Content-Based Methods

Content-based algorithms recommend items similar to those a user has preferred in the past by matching item features (e.g., keywords, categories) [52–54]. While effective for addressing the item cold-start problem, they rely heavily on manual feature engineering and often struggle to capture the complex semantic interactions between different modalities (e.g., connecting visual style with textual sentiment) [47, 55].

2.2.2 Collaborative Filtering (CF)

CF remains the cornerstone of personalized recommendation, operating on the assumption that users with similar behaviors exhibit similar preferences. It is broadly categorized into two types [56, 57].

Memory-based CF: This approach, including user-based and item-based CF, directly computes similarity between users or items using interaction vectors (e.g., via cosine or pearson correlation) [58–62]. While intuitive, memory-based methods suffer significantly from data sparsity and lack the ability to model transitive relationships.

Model-based CF: To address scalability and sparsity, model-based methods employ machine learning algorithms to learn predictive models [63–65]. (1) Matrix factorization (MF): MF decomposes the interaction matrix into latent user and item factors [66, 67]. Variants like PMF and SVD++ integrate probabilistic models or implicit feedback to enhance accuracy [68–70]. (2) Clustering-based models: These methods group similar users or items into clusters (using algorithms like K-means or Fuzzy C-means) to reduce dimensionality and smooth out noise [71–74].

However, despite their success, traditional CF models typically process interactions as independent linear events. They lack the architectural depth to encode the high-order structural dependencies (e.g., A-B-C-D paths) that are inherent in social or knowledge graphs [75, 76].

2.3 Limitations in Multi-type Scenarios

The transition from traditional methods to GNNs is driven by the inability of CF and content-based models to handle complex heterogeneity [77–83]: (1) Inability to model behavioral hierarchies. Traditional methods typically flatten multi-behavior data into a single matrix or treat them independently, failing to capture the cascading dependencies (e.g., view → cart → purchase). (2) Shallow feature integration. They lack mechanisms to align heterogeneous modalities into a shared semantic space, often treating images and text as static feature vectors rather than interactive signals. (3) Lack of high-order relational reasoning. Most importantly, traditional methods struggle to model multi-hop connectivity. They cannot effectively propagate information across complex multi-relational structures (e.g., user-social-user-item paths), which limits their reasoning capability in knowledge-aware or social recommendation.

3 Multi-types Graph Neural Networks-Based RS

Graph neural networks (GNNs) have become a core technique for modeling complex user-item interaction graphs in modern recommender systems [84–86]. Compared with traditional methods, GNNs can automatically learn high-order collaborative signals and rich structural patterns from multi-source heterogeneous data, alleviating manual feature engineering and improving robustness to sparsity and cold-start issues [87–90].

Following the multi-type taxonomy in Fig. 1, this section conducts a comparative analysis through three heterogeneity dimensions [91–94]: multi-relational, multi-behavior, and multi-modal. We also discuss spatio-temporal dynamics as a special case of dynamic multi-relational modeling. Instead of organizing the review by GNN families (GCN, GraphSAGE, GAT, GAE, and GSTN), we emphasize how different methods address each type dimension, the trade-offs they introduce, and the gaps that remain.

3.1 Handling Behavioral Heterogeneity (Multi-Behavior)

Real-world interactions are multifaceted (e.g., click, cart, purchase). The primary challenge here is semantic disentanglement: differentiating the signal strength of various behaviors and modeling their cascading dependencies without introducing noise. To enable comparison, we evaluate methods along four aspects: behavior dependency modeling, behavior imbalance and noise handling, objective alignment with ranking, and computational scalability [95–98].

3.1.1 Cascading and Multiplex Modeling

GCN-based models typically construct heterogeneous graphs where different behaviors form distinct edges [99, 100]. To capture the hierarchical nature of interactions, MB-CGCN [85] employs a cascading design where embeddings from low-signal behaviors (e.g., clicks) serve as inputs for high-signal behaviors (e.g., purchases). By stacking LightGCN [91] modules in a chain (click → cart → purchase), it explicitly models the conversion probability. Similarly, MGNN [101] and MBGCN [19] utilize multiplex graph structures to learn behavior-specific embeddings. Li et al. [100] further proposed a fusion model based on GCN to combine user heterogeneous characteristics with temporal behavior features.

Critical analysis (uniformity bias): While effective at structural aggregation, standard GCNs suffer from a uniformity bias [102–106]. Their aggregation mechanism typically treats a ‘purchase’ edge and a ‘click’ edge

with fixed normalization, struggling to automatically distinguish the varying semantic intensity of these interactions. This often necessitates manual graph engineering (e.g., separate subgraphs) to prevent sparse high-value signals from being drowned out by abundant low-value clicks, as observed in the comparative baselines of recent works (Fig. 2 for a summary of GCN applications). Cascading or multiplex designs help, but GCN usually needs explicit behavior separation to avoid noisy low-value interactions dominating the signal [107–111].

3.1.2 Attention-Based Disentanglement

GAT architectures address the uniformity limitation by introducing learnable attention weights [112]. MutualRec [84] employed GAT to differentiate the contribution of implicit feedback versus explicit ratings. By assigning higher attention scores to critical behaviors, GAT acts as a soft-selection filter. Chen et al. [113] and Wang et al. [114] also utilize attention to intuitively allocate weights between neighbor nodes and side information, effectively integrating heterogeneous behavioral types into node representations [115, 116].

Critical analysis (superior disentanglement): GAT is arguably the most suitable architecture for multi-behavior types [107, 117–121]. It acts as a soft-selection filter, automatically learning to prioritize ‘cart’ or ‘purchase’ signals over noisy ‘clicks’ during message passing. This offers a significant advantage over GCN’s static aggregation in noisy multi-behavior environments (Fig. 3 for GAT trends). Attention offers stronger end-to-end behavior disentanglement, but practical use still depends on careful regularization in sparse settings [122–124].

3.1.3 Implicit Sampling

Some approaches like MRIG [125] utilize GraphSAGE to sample neighbors from multi-relational item graphs involving target and auxiliary behaviors.

Critical analysis (sampling bias): While scalable, GraphSAGE’s random sampling poses a risk in this context [126–129]. High-value behaviors are often sparse; uniform sampling may statistically miss these critical interaction partners, leading to a loss of key user intent signals compared to full-graph GCN or attention-based weighting. Sampling improves scalability, yet multi-behavior learning often requires behavior-aware or importance-aware sampling to preserve rare but valuable signals.

3.1.4 Latent Reconstruction

GAE-based methods optimize embeddings by reconstructing the interaction graph [130]. *Critical analysis (ranking gap):* GAEs face a misalignment in multi-behavior tasks. Their objective is often link prediction (existence) rather than ranking (preference order), making them less direct than GCN/GAT for distinguishing the relative

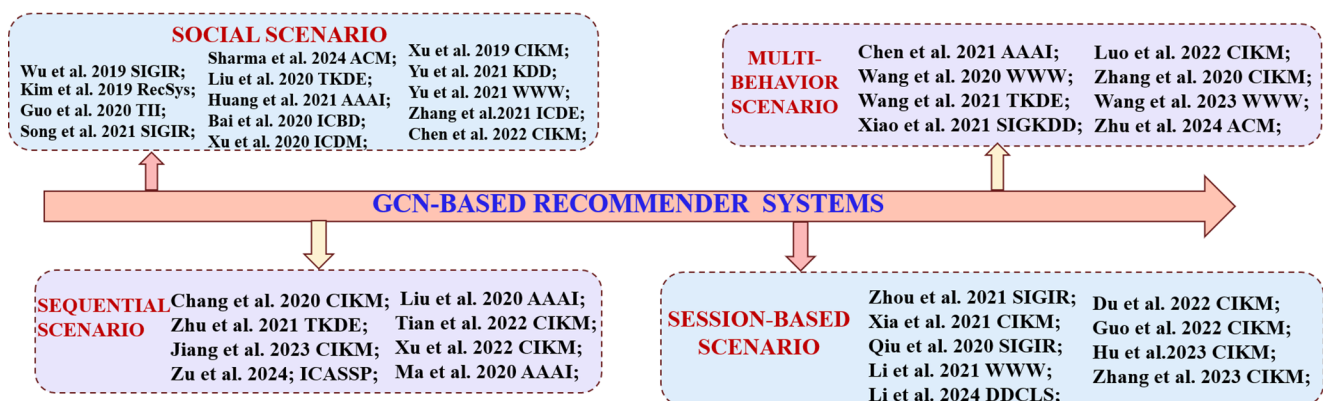


Fig. 2 Summary of GCN-based literature in different scenarios in top-tier conferences

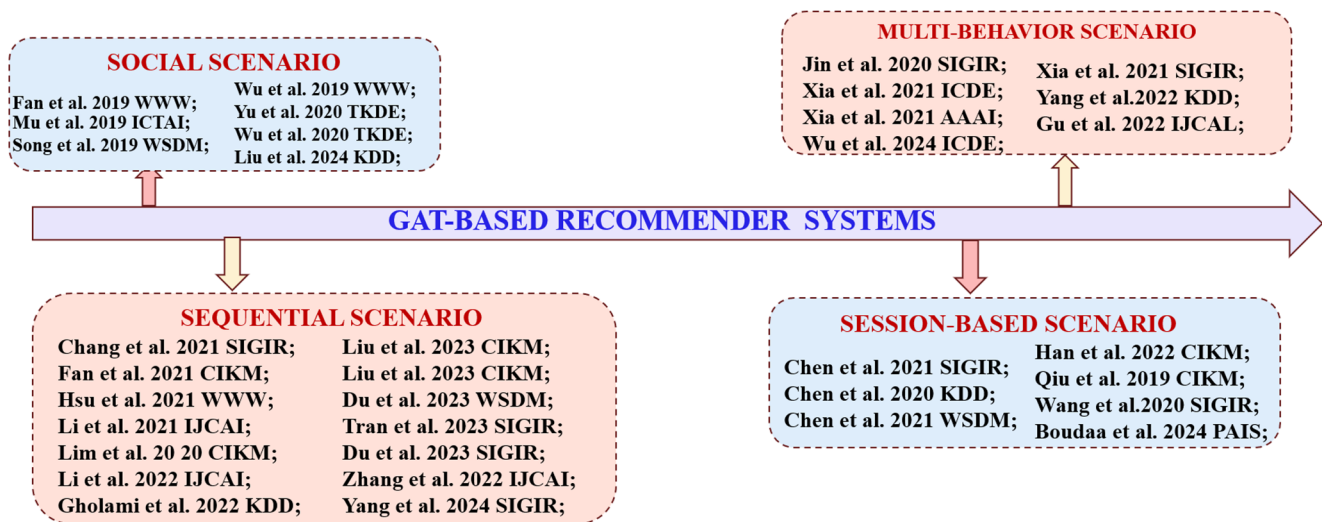


Fig. 3 Summary of GAT-based literature in different scenarios in top-tier conferences

importance of ‘buy’ vs ‘view’. Reconstruction objectives can complement behavior modeling, but ranking-aligned losses are typically necessary to prioritize high-value behaviors [131–135].

3.2 Integrating Modal Attributes (Multi-modal)

The core challenge in multi-modal RS is feature alignment: bridging the semantic gap between static content features (visual, textual) and dynamic collaborative signals. We compare methods by where fusion is performed (input, message passing, or objective), how cross-modal alignment and denoising are achieved, their cold-start capability for new items, and their additional computational cost [136, 137].

3.2.1 Inductive Fusion

GraphSAGE is particularly dominant in multi-modal settings. Ying et al. [138] successfully applied GraphSAGE to web-scale recommender systems, generating embeddings for billions of items by aggregating their rich content features. Its inductive nature allows it to compute embeddings for new nodes (cold-start) by aggregating their content features immediately [138, 139].

Critical analysis (inductive strength): GraphSAGE excels here because it treats modal features as dynamic inputs for the aggregator function. Unlike transductive GCNs that require retraining for new ID embeddings, GraphSAGE effectively bridges the gap between content modalities and graph structures for unseen items. Inductive aggregation makes GraphSAGE attractive for cold-start multi-modal items, especially when content features are informative and frequently updated [140, 141].

3.2.2 Latent Alignment

GAEs offer a unique advantage by using dual decoders. Models like SMVAE-UP [18] and CWAE [142] reconstruct both the graph structure and the node attributes. HybridGNN-SR [143] combines unsupervised variational autoencoders with supervised routing to represent item transition patterns alongside content features.

Critical analysis (denoising): This generative process forces the learned embeddings to capture the intrinsic correlation between valid modal signals and interaction patterns, effectively filtering out modal noise (e.g., irrelevant visual background) during the encoding phase [144–147]. Generative reconstruction can improve modal robustness, but training cost and objective mismatch should be considered for top-K recommendation [148, 149].

3.2.3 Static Limitations

Standard GCNs typically treat multi-modal features as fixed initializations for nodes. They lack an intrinsic mechanism to dynamically re-weight conflicting modalities (e.g., when visual features contradict textual descriptions) during propagation, often requiring auxiliary modules to handle modal misalignment [150, 151]. When modalities conflict or drift over time, GCN-style static initialization is usually insufficient without additional alignment or attention modules.

3.3 Modeling Structural Complexity (Multi-relational and Spatio-temporal)

This dimension involves complex topologies such as social networks, knowledge graphs, and spatio-temporal dependencies. We analyze methods by relation encoding (type-specific propagation), hierarchy or compositional reasoning, scalability on large heterogeneous graphs, and the ability to incorporate temporal dynamics without prohibitive overhead [152–158].

3.3.1 Social and Knowledge Relations

Global structural propagation: GCN. GCNs are the backbone for multi-relational tasks [159, 160]. In social recommendation, DiffNet [161] and HGNN [162] leverage GCNs to diffuse social influence across user-user links, effectively alleviating data sparsity by borrowing information from high-order social neighbors. Yu et al. [163] and Xu et al. [164] further incorporated semantic-aware meta-paths and adversarial training to enhance robustness. Hete-Graph [165] demonstrated GCN's capability in processing complex heterogeneous graphs for diversified recommendation.

Critical analysis (structural efficiency). GCNs are inherently structurally aware. By explicitly modeling relation-specific adjacency matrices (as in R-GCN), they efficiently propagate dependencies across heterogeneous entities (users, items, entities) without the high computational overhead of attention mechanisms [166, 167]. For large heterogeneous graphs, type-specific GCN variants provide a strong efficiency-expressiveness balance.

Scalability bottlenecks: GAT. GAT effectively models social influence weights. [14] analyzed dynamic social influence, while ASR [15] and DANSER [16] utilized hierarchical and dual-attention mechanisms to capture homophily and influence effects.

Critical analysis (complexity). While GAT offers superior expressiveness for heterogeneous relations, calculating attention coefficients for every edge type in a large-scale knowledge graph leads to quadratic complexity. This often necessitates graph pruning that may lose structural information compared to GCN's efficient matrix operations. GAT improves relation weighting, but its cost grows quickly with graph size and relation types, so pruning or sparse attention is often required [168–170].

3.3.2 Sequential and Spatio-Temporal Dynamics

Sequential modeling (GCN/GAE). To capture sequential patterns, ISSR [171] and Guo et al. [92] applied GCN to user-item bipartite graphs linked over time or across domains. Shafqat et al. [172] and Yin and Zheng (2021) [173] simplified GCN training for session graphs. HybridGNN-SR [143] combined unsupervised GAE with supervised learning to represent item transition patterns in sessions.

Spatio-temporal specialization (GSTN). For spatio-temporal relations (a special class of multi-relational data), GSTN is the specialized solution [174–177]. Models like STGCN [178] and ASTGCN [179] capture periodic patterns and sequential dependencies in POI recommendation that static architectures fail to encode. [180, 181] utilized a generative adversarial network framework to learn style features in dynamic movements.

Critical analysis (complexity trade-off). The specialization of GSTN comes at a cost. Modeling dynamic dependencies (time+space) is computationally expensive [182–184]. Consequently, existing GSTN approaches

often sacrifice the richness of other types (e.g., ignoring complex multi-behavior interactions) to maintain temporal computational feasibility. Spatio-temporal modeling yields richer dynamics [185, 186], but most GSTN-based RS must simplify other types to remain tractable.

3.4 Comparative Synthesis: Matching GNN Architectures to Multi-type Challenges

While the previous sections reviewed architectures individually, selecting the optimal GNN backbone requires understanding their trade-offs regarding specific multi-type challenges [187, 188]. Table 3 summarizes these comparative capabilities.

- **Handling behavioral heterogeneity (multi-behavior).** GAT and its variants outperform GCN and GraphSAGE in explicitly modeling the semantic gaps between behaviors [189]. GCNs require predefined multi-view graph construction (manual separation of behaviors) to achieve similar results, whereas GAT learns these distinctions end-to-end. However, when behavior data is extremely sparse, GAE-based methods often show superior robustness by reconstructing missing interactions rather than purely aggregating them.
- **Integrating modal attributes (multi-modal).** GraphSAGE offers the best flexibility for multi-modal integration in dynamic environments due to its inductive attribute aggregation. In contrast, GCN-based approaches often treat modal features as static initialization, failing to capture the interaction between modality updates and topological changes. For complex modal alignment, GAT provides finer granularity but at a higher computational cost.
- **Modeling structural complexity (multi-relational).** For complex multi-relational structures like knowledge graphs, GCN variants (e.g., R-GCN) provide the best balance between structural expressiveness and computational efficiency. GAT models suffer from scalability issues here, while GraphSAGE may oversimplify relational semantics through sampling.
- **Handling spatio-temporal dynamics (multi-relational/dynamic).** Distinct from the above, GSTN is the only architecture specialized for spatiotemporal relations. While it excels in capturing time-evolving patterns (e.g., in POI recommendation), it struggles to integrate efficiently with complex multi-behavior or heavy multi-modal data due to the high computational cost of modeling dynamic dependencies.

In summary, the choice of architecture should be dictated by the dominant heterogeneity type. GAT for semantic heterogeneity (multi-behavior), GraphSAGE for attribute/modal dynamism (multi-modal), and GCN for structural complexity (multi-relational), and GSTN for temporal evolution constraints.

3.5 Summary and Quantitative Analysis

3.5.1 Dominance of GCN and GAT

As indicated in Table 2, GCN and GAT architectures remain the dominant backbones in the field. GCNs act as the structural foundation, often integrated with other paradigms (e.g., GAE for implicit factor mining or GSTN for adversarial generation) to enhance robustness. While GraphSAGE excels in scalability for POI tasks, its standard sampling mechanism often overlooks the semantic granularity required for complex behavior modeling. Similarly, GSTN is indispensable for spatiotemporal tasks but suffers from high computational memory costs, limiting its adoption in general-purpose recommendation.

3.5.2 Evolution of multi-behavior modeling

Strategies for modeling auxiliary behaviors have evolved from simple aggregation to complex semantic disentanglement. Early approaches [125] treated all behaviors in a unified graph, performing uniform convolution. Subsequent methods like [114] and [113] introduced subgraph learning and intuitive allocation to separate behavior

Table 2 Studies on strategies for GNN methods

Literature	Description	Contributions	Limitations
Ying et al. [138] Guo and Wang [205] Chen et al. [192] Gmati et al. [185] Tran et al. [177] Zhang et al. [195]	GCN, applied to Web recommendation tasks This approach works for all recommendation tasks Applied to large-scale heterogeneity Recommendation tasks for graphs	Improve the robustness of the model Reduce processing latency Handling heterogeneous graph data	Cannot effectively learn from other large-scale graph representations The memory access model is complex It only applies to two entities: users and items
Shafqat et al. [172] Liu et al. [206]	Online product recommendation task	The arbitrary neighbor sampling in the GCN model is simplified, which improves training efficiency and reduces computational complexity and time	Session graphs need to be formed and are not adapted to all RS Scenario
Bonet et al. [207] Deldjoo et al. [208]	Big Data recommendation tasks	Improve the performance and accuracy of RS	Unable to address cold-start and data sparsity issues; overlooks the interpretability of RS
Wu et al. [187] Xu et al. [196] Xu et al. [164] Deng et al. [181]	Generation Adversarial network Online Video recommendation	Enhance the stability of recommender systems Improve the accuracy of recommendations	Only useful for content-based recommendations like reviews Difficult to model long sequences in large graphs; requires high-quality information
Jiang et al. [15] Wu et al. [187] Xiao et al. [96] Dang et al. [96] Li et al. [100] Salamat et al. [202] Sang et al. [17]	GCN, social recommender Rating prediction task, TOP-K recommendation task Graph Attention Neural Networks	It can find the potential social transmission utility and improve the accuracy Solve the problem of data sparsity and cold start It can automatically capture rich semantic information and complex implicit relationships between users and items	The model is complex and needs to extract enough information data Longer running time The sequential nature of interactions between users is not considered
Zheng et al. [203] Yao et al. [142] Deng et al. [204] Ohtomo et al. [18]	GCN, social recommender Graph autoencoders	Capture data correlations to improve recommendation performance Personalized recommendation information for each user from a large amount of information	The time order factor is not considered Long training time
Park et al. [180] Han et al. [193]	GCN, Sports Style Recommendation GCN, POI recommendation	It can extract features in both spatial and temporal dimensions Alleviate data sparsity issues	The algorithm is very sensitive to random noise Ignores spatio-temporal sequences and contextual information between nodes

representations. More recent advancements, [190] utilize attention mechanisms and knowledge-aware injections to capture the cascading influence of behaviors (e.g., click → cart → purchase). However, a persistent challenge is effectively distinguishing the semantic intensity of different behaviors (e.g., the strong intent of “cart” vs. the weak signal of “click”) without introducing excessive noise.

3.5.3 Critical Analysis of Multi-type Support (Fig. 4)

To provide a data-driven perspective on the field’s maturity, Fig. 4 presents a heatmap summarizing representative models across three multi-type dimensions. The visualization reveals three critical insights:

- *Saturation in multi-relational modeling.* As evidenced by the continuous yellow blocks (value=1.0) in the “multi-relational” row, structural heterogeneity is the most mature area. Almost all GNN-based RS, regardless of their specific focus, inherently model user-item or item-entity relations, confirming that graph structure is the fundamental enabler of modern RS.
- *The multi-modal gap.* A significant “cold spot” (purple/value=0.0) appears in the “multi-modal” row. With the exception of MMGCN, most multi-behavior models (e.g., MBGCN, MB-CGCN) completely ignore modal

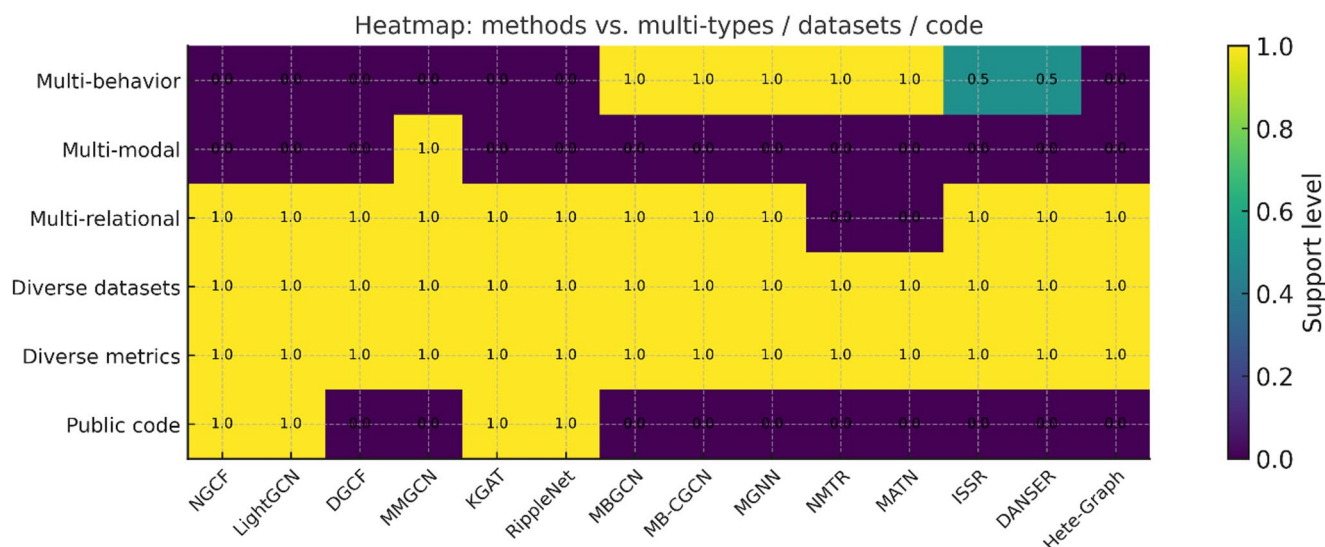


Fig. 4 Heatmap summarizing representative GNN-based recommendation methods over multi-behavior, multi-modal, multi-relational support, datasets, metrics, and code availability (0/0.5/1 denote increasing support levels)

attributes like text or images. This highlights a critical research gap: existing methods struggle to unify behavioral heterogeneity (actions) with modal heterogeneity (content), representing a prime opportunity for future work.

- *The reproducibility crisis in multi-behavior research.* A concerning pattern emerges in the “public code” row. While general baselines like LightGCN and NGCF exhibit high reproducibility (value=1.0), the specialized multi-behavior cluster (from MBGCN to MATN) shows a distinct lack of open-source implementations (value=0.0). This lack of code availability hinders comparative benchmarking and slows down the iteration of multi-behavior algorithms compared to general GNN architectures.

4 Problems and Perspectives

Graph Neural Networks (GNNs) have shown significant promise in the domain of recommender systems (RS) due to their ability to capture complex user-item relationships and exploit graph structures effectively. However, as summarized in Table 4, substantial challenges remain, especially when GNNs are deployed in multi-behavior, multi-modal, and multi-relational contexts. In this section, we outline major open problems and link them to recent literature and emerging solutions, followed by forward-looking research directions.

4.1 Open Problems

Overall, multi-types analysis for GNN-based RS has the following common open problems [191–193]. As for each graph-based multi-behavior recommendation algorithm, such as Graph spatial-temporal networks (GSTN) (5), Graph Auto Encoders (GAE) (6), Graph Attention Networks (GAT) (7), Graph Sample and Aggregate (GraphSAGE) (8), and Graph Convolution Networks (GCN) (10) recommendation all suffer from various open problems, as shown in Table 4.

4.2 Scalability and Efficiency

With the rapid growth of user-item interactions, real-world RS graphs are becoming increasingly large and complex. Training GNNs on such large-scale graphs is computationally expensive in terms of both time and memory.

- **Challenges:** (1) Designing scalable message-passing mechanisms that maintain performance on massive multi-relational graphs; (2) Developing efficient sampling and pruning strategies for mini-batch training to reduce the cost of iterative neighbor aggregation.
- **Emerging solutions:** Recent works on GraphSAGE-like neighbor sampling, redundancy-aware subgraph sampling, and distributed GNN training (e.g., for web-scale recommendation in Table 3) provide partial solutions, but more general frameworks that jointly consider multi-type heterogeneity and system constraints are still lacking.

4.3 Over-Smoothing in Deep GNNs

As the number of GNN layers increases, node embeddings tend to become indistinguishable, leading to over-smoothing, especially in dense multi-relational graphs.

- **Challenges:** (1) Balancing the depth of GNNs to maintain expressiveness without losing feature diversity; (2) Designing architectures that capture long-range dependencies without over-smoothing.
- **Emerging solutions:** Residual connections, jumping knowledge, normalization layers, and layer-wise contrastive regularization have been explored in recent GNN studies, but their effectiveness in multi-behavior and multi-modal RS is still under-explored and calls for systematic evaluation.

4.4 Cold-Start Problem

Handling new users, items, or interaction types remains a fundamental issue. GNNs often rely heavily on existing graph structures, which may not be sufficient in cold-start scenarios.

- **Challenges:** (1) Incorporating external features such as textual content, images, and metadata into GNN-based models in a principled manner; (2) Designing hybrid approaches that leverage pre-trained embeddings, meta-learning, or transfer learning across domains and behavior types.

Table 3 Comparative suitability of GNN architectures for multi-type RS

Architecture	Suitability for multi-behavior	Suitability for multi-modal	Suitability for multi-relational	Key trade-off
GCN	Low (Requires separate graphs)	Medium (Static features)	High (Efficient structural prop.)	Efficiency vs. Semantic distinguishing
GraphSAGE	Low (Sampling bias risk)	High (Inductive attribute handling)	Medium (May lose edge semantics)	Scalability vs. Signal loss
GAT	High (Learnable weights)	High (Adaptive fusion)	Low (Scalability issues)	Expressiveness vs. Computation Cost
GAE	Medium (Reconstruction)	Medium	Medium	Latent modeling vs. Direct ranking
GSTN	Low (Focuses on sequential/temporal)	Low (Seldom combined with modality)	Medium (Specialized in Spatio-Temporal)	Dynamics modeling vs. Complexity

High indicates that the architecture can natively model the target type with end to end learning and has multiple representative RS studies; Medium indicates that the type is supported with additional modules or careful graph construction; Low indicates that the architecture typically requires manual decomposition or does not scale or align well with the target type [209, 210]

Table 4 Open problems of DL-based RS

GNN algorithms	Recommendation scenarios	Literatures	Open problems
GCN	Sequential recommendation	Guo et al. [211]	Cold start, sparsity data
	Social recommendation	Zhao et al. [212]	Heterogeneous diagram, interpretability, dynamic behavior
	Social recommendation	Yu et al. [163]	Data sparseness, personalized questions
	Multi-behavior recommendation	Xu et al. [196]	Handling sparse relationships, and adapting to dynamic relational changes
	Sequential recommendation	Lan et al. [200]	Diversity vs. Accuracy
	Cross-domain recommendation	Liu et al. [213]	Neighborhood polymerization
	Cross-domain recommendation	Zhang et al. [214]	Information efficiency
	Multi-behavior recommendation	Ma et al. [215]	Heterogeneous Graphs, Diversity, and Accuracy
GAT	Sequential recommendation	Tao et al. [216]	Building dynamic graph and item information
	Cross-domain recommendation	Liu et al. [217]	data sparsity
	Sequential recommendation	Wang et al. [218]	Higher-order relational modeling, interpretability
	Cross-domain recommendation	Subaranjani et al. [219]	Cold-start and explainability
	Social recommendation	Jiang et al. [15]	Dynamic representation problem
	Social recommendation	Salamat et al. [202]	Big data, heterogeneous diagram, interpretability, dynamic behavior
	Social recommendation	Tu et al. [220]	Data sparsity, personalization problem, knowledge graph
	Multi-behavior recommendation	Xia et al. [221]	Graph Complexity
GraphSAGE	Session recommendation	El et al. [222]	Cold start
	Session recommendation	Gu et al. [223]	Heterogeneous graph, potential information
	Multi-behavior recommendation	Yang et al. [224]	Capture useful information
	Session recommendation	Bansal et al. [225]	Data sparsity, cold start
GAE	Multi-behavior recommendation	Zhang et al. [195]	Computational efficiency and resource consumption, and complexity in multi-behavior interaction modeling
	Sequential recommendation	Salha et al. [226]	Accuracy, personalization
GSTN	Social recommendation	Xie et al. [227]	The diversity of the recommended
	Cross-domain recommendation	Guo et al. [94]	algorithmic complexity
	Cross-domain recommendation	Budhiraja [199]	Applicability of the algorithm
	Sequential recommendation	Chang et al. [93]	Not consider noisy historical sequences from multiple types of behaviors
	Social recommendation	Liu et al. [198]	Big data, relationship dynamic change problem
	Multi-behavior recommendation	Haque et al. [197]	Behavior interaction modeling
	Multi-behavior recommendation	Xia et al. [228]	Heterogeneous relationships under multiple types
	Multi-behavior recommendation	Cheng et al. [85]	Semantic Capture Problem

- Emerging solutions: Recent GNN-based RS models have started to use knowledge graphs, multi-modal encoders, and meta-learning to alleviate cold-start (Table 4 for cross-domain and knowledge-aware settings), but unified multi-type cold-start solutions remain an open problem.

4.5 Multi-behavior Interaction Modeling

Real-world RS involve multiple types of user-item interactions (e.g., clicks, adds-to-cart, purchases, reviews). Many GNN models still focus on single-interaction types or treat multiple behaviors in a simplistic manner.

- Challenges: (1) Modeling and fusing multi-behavior data effectively while capturing their hierarchical and causal relationships; (2) Differentiating the importance of various behaviors in the recommendation process under different tasks and domains.
- Emerging solutions: Cascading GCNs, multiplex graphs, attention-based fusion, and disentangled representation learning have been proposed for multi-behavior recommendation (e.g.,) [19, 85, 101, 194–197] in Table 4),

but there is still a lack of unified multi-type frameworks that can jointly handle multi-behavior, multi-modal, and multi-relational signals.

4.6 Dynamic and Temporal Graphs

User preferences and item characteristics evolve over time, leading to dynamic graph structures. Static GNNs often fail to capture such temporal dynamics.

- Challenges: (1) Designing GNNs that can efficiently adapt to graph changes (e.g., streaming updates, evolving relations); (2) Incorporating temporal dependencies and periodic patterns to predict evolving user–item interactions.
- Emerging solutions: GSTN-based models, temporal GNNs, and sequential/session-based GNN-RS have shown promise in spatio-temporal and POI scenarios ([93, 94, 198–200] in Table 4). However, integrating temporal modeling with rich multi-behavior and multi-modal information remains an open research direction.

4.7 Explainability

GNN-based RS often function as black-box models, making it difficult to interpret the reasoning behind their recommendations, which raises concerns about transparency, fairness, and user trust.

- Challenges: (1) Developing interpretable GNN architectures or post-hoc explanation methods that can highlight influential neighbors, behaviors, and modalities; (2) Balancing recommendation accuracy with explainability, fairness, and robustness.
- Emerging solutions: Attention-based models, path-based explanations (e.g., via meta-paths or subgraphs), and counterfactual analysis have been explored in recent work, but systematic explainable multi-type GNN-RS, especially for multi-behavior and knowledge-aware graphs are still in their infancy.

4.8 Perspectives

RS provide substantial convenience for users and platforms, and many personalized algorithms have shown strong performance in diverse application domains. However, the continuous growth of multi-type data (video, news, e-commerce, social media, POI, etc.) and the rapid evolution of GNN architectures create new opportunities and challenges. Building on the above open problems and recent literature (summarized in Tables 3 and 4), we outline several promising research directions:

- 1) Task-aware graph structure learning for multi-type recommendation. Most current graph-based RS construct graphs directly from observed user histories, implicitly treating unobserved items as negative or neutral, which may be biased by exposure mechanisms, long-tail effects, or platform policies. Future work should investigate task-aware and uncertainty-aware graph structure learning, where edge construction, pruning, and reweighting are jointly optimized with the recommendation objective. For multi-type RS, this includes learning which behaviors, relations, and modalities should be connected or emphasized under different tasks (e.g., ranking, diversification, fairness).
- 2) Beyond static KG: dynamic and semantic knowledge integration. Knowledge graphs have been shown to mitigate cold-start and sparsity by enriching item and user representations, extending connectivity, and improving diversity and interpretability. However, most existing KG-based RS rely on static knowledge and hand-crafted relations. Future research should explore dynamic, semantic knowledge integration, including: (i) automatically constructing and updating knowledge graphs from user-item interactions and content; (ii) learning

semantic relations (e.g., intent, style, sentiment) from multi-modal data; and (iii) modeling how these knowledge structures evolve over time in multi-type GNN-based RS.

3) Integrating GNNs with large language models (LLMs) and generative paradigms. Recent progress in large language models (LLMs) and foundation models opens new possibilities for GNN-based recommendation. Potential directions include:

- 4) Using LLMs to generate or refine textual profiles for users/items and to provide natural language rationales for graph-based recommendations.
- 5) Employing LLMs as controllers or priors for graph construction and edge typing, especially in multi-modal and knowledge-rich environments.
- 6) Designing GNN+LLM hybrid architectures where GNNs capture high-order collaborative signals on graphs and LLMs provide semantic reasoning and instruction-following capabilities (e.g., conversational or explainable recommendations).
- 7) These directions can help address cold-start, improve interpretability, and enable more flexible interaction forms, but raise new challenges in efficiency, alignment, and evaluation.

(4) Explainable and trustworthy multi-type GNN-based RS. As multi-type GNN-based RS become more complex, user trust and regulatory requirements make explainability and robustness increasingly important. Future work should:

- Develop intrinsically interpretable multi-type GNN architectures (e.g., prototype-based, rule-guided, or causal-graph-inspired models) that can provide behavior-level and relation-level explanations;
- Combine GNNs with explainable AI (XAI) techniques, such as subgraph rationales, attention visualization, and counterfactual reasoning, tailored to multi-behavior and multi-relational graphs;
- Incorporate fairness, robustness, and privacy constraints into multi-type GNN-RS, moving from “black-box accuracy” to trustworthy recommendation in realistic scenarios.

In summary, while existing GNN-based RS have achieved notable progress in handling various types of heterogeneous information, many open problems remain when moving toward unified multi-type modeling. Bridging GNNs with emerging paradigms such as LLMs and explainable AI, and jointly addressing scalability, dynamics, and trustworthiness, will be key to advancing the next generation of multi-type GNN-based recommender systems.

5 Conclusion

This study has systematically reviewed collaborative filtering (CF)-based personalized recommendation algorithms, with a particular emphasis on model-based approaches such as clustering- and matrix factorization-based methods. By analyzing their strengths and limitations, we highlighted how recent research integrates CF with graph neural networks (GNNs), data mining, and auxiliary information to address long-standing challenges including cold start, data sparsity, scalability, diversity, and interpretability.

A key observation is that multi-behavior data offers significant potential to enhance clustering efficiency, improve customer segmentation accuracy, and boost recommendation performance. Combining multiple user behaviors with advanced graph construction techniques in GNN-based recommendation systems emerges as a promising pathway to overcoming accuracy bottlenecks and mitigating overfitting. Such integration not only improves platform recommendation performance but also better satisfies users’ personalized demands.

Looking forward, future research should explore richer multi-type and multi-modal data sources, incorporate dynamic and real-time graph learning to adapt to evolving user preferences, and develop scalable architectures for large-scale recommendation scenarios. Additionally, integrating GNNs with emerging paradigms such as large language models could further advance the interpretability and adaptability of recommender systems.

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Data Availability The manuscript has no associated data.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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